I Remember That Question: Using Eye Tracking to See How Prequestions Shape Reading

Jason Geller

Department of Psychology and Neuroscience, Boston College

Author Note

Jason Geller https://orcid.org/0000-0002-7459-4505

All data (raw and processed), materials, analysis scripts and code, and the source of this manuscript are available at https://github.com/jgeller112/Prequestions_EyeTracking. The author has no conflicts of interest to disclose. I thank Shana Carpenter for help with the materials and with conceptualizing the study.

Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: *Jason Geller***:** conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing – original draft, writing – review & editing, visualization, and project administration

Correspondence concerning this article should be addressed to Jason Geller, Department of Psychology and Neuroscience, Boston College, Mcguinn Hall 405, Chestnut Hill, MA 02467-9991, USA, Email: drjasongeller@gmail.com

Abstract

Answering questions before learning (prequestions) improves memory for the information they address, but how it does so is unclear. A common proposal is that prequestions direct attention to that information. We tested this account by recording eye movements while participants read a text about polio and tuberculosis after answering prequestions about one topic or none, so that every target sentence was read both with and without prequestions. Prequestions improved memory only for the prequestioned information, and eye movements showed the same specificity. Ex-Gaussian models, which separate typical reading time from occasional unusually long readings, showed that prequestions did not change typical reading of the answer sentences. Instead, these sentences received more unusually long readings, were looked back to more often, and were re-read for longer; other sentences were unaffected. Prequestions thus guide attention selectively rather than heightening it, consistent with additional processing when readers recognize an answer.

Keywords: prequestions, pretesting, eye tracking, reading, attention, learning

I Remember That Question: Using Eye Tracking to See How Prequestions Shape Reading

A common study strategy involves retrieving information from memory by answering questions after a learning event (Hartwig & Dunlosky, 2012). This technique, known as *retrieval practice*, has been shown to improve long-term retention across a wide range of materials and learning contexts, a phenomenon known as the *testing effect* (S. K. Carpenter, 2012; Dunlosky et al., 2013; Kornell & Vaughn, 2016; Roediger & Karpicke, 2006; Rowland, 2014). Surprisingly, answering questions *before* a learning event can also improve subsequent memory, even when learners have little prior knowledge of the material and most of their initial answers are incorrect. This phenomenon, known as the *prequestion effect*, is particularly intriguing because, unlike traditional retrieval practice, it does not require learners to retrieve previously studied information. Although this finding is consistent with the broader literature suggesting that effortful and even error-prone learning experiences can benefit long-term retention (Bjork, 1994), why prequestions improve learning remains unclear.

The benefits of prequestions are typically demonstrated in experiments in which learners are assigned to a prequestion group or a control group (S. K. Carpenter et al., 2023). Before encountering the learning material, the prequestion group attempts to answer questions about the upcoming content, often guessing when they do not know the answers, whereas the control group encounters the material without answering any questions. After learning, both groups complete a final memory test assessing information that was targeted by the prequestions as well as information that was not. The typical finding is that learners who received prequestions remember the targeted information better than those who did not, a phenomenon known as the specific benefit of prequestions (Pan & Carpenter, 2023). In some cases, prequestions also improve memory for information that was not directly targeted, known as a general benefit of prequestions. However, whereas the specific benefit is robust, the general benefit is less consistent. Two recent meta-analyses found substantial specific benefits ($g = 0.54$, St. Hilaire et

al., 2024; g = 0.66, King-Shepard et al., 2025), but no reliable general benefit (g = 0.04, St. Hilaire et al., 2024; g = 0.01, King-Shepard et al., 2025). Nevertheless, some studies have reported general benefits depending on the learning materials (S. K. Carpenter & Toftness, 2017), the types of questions used (Bostan & Ozcelik, 2026; St. Hilaire et al., 2019), and the placement of non-prequestioned information relative to prequestioned content (Sana & Carpenter, 2023). The specific benefit, by contrast, has been demonstrated acorss a wide variety of study materials and settings (S. K. Carpenter et al., 2018; S. K. Carpenter & Toftness, 2017; Geller et al., 2017; Richland et al., 2009; Soderstrom & Bjork, 2023; St. Hilaire & Carpenter, 2020; see Pan & Carpenter, 2023 for a review).

Why do preqeustions improve learning? Several non-mutually exclusive mechanisms have been proposed (King-Shepard et al., 2025; Pan & Carpenter, 2023). Prequestions may stimulate curiosity about the answers (Berlyne, 1954, 1962), encourage elaboration by activating related prior knowledge (Kornell et al., 2009), or provide a metacognitive "reality check" that makes learners aware of gaps in their knowledge (Bjork et al., 2013). Although these explanations differ, they share the possibility that prequestions direct learners' attention toward information relevant to the questions (King-Shepard et al., 2025). According to an *attention-based account*, prequestions signal which information is important, encouraging learners to attend to and process that information more thoroughly when they encounter it (Pan & Carpenter, 2023). Consistent with this account, researchers looking at eye movements during a learning episode have found that learners spend more time looking at (Bostan & Ozcelik, 2026; Yang et al., 2021), re-inspecting, and returning to prequestioned information after reading subsequent material (Lewis & Mensink, 2012).

Although these findings support an attention-based account, several results suggest that directing attention toward prequestioned information may not, by itself, fully explain the benefits of prequestions. If prequestions primarily function by signaling which information deserves

attention, other methods of highlighting that information might be expected to produce similar benefits. However, answering prequestions has been found to improve memory more than studying material with the tested concepts emphasized or reading questions without answering them (Richland et al., 2009), and more than reading learning objectives or statements of the target facts (Sana et al., 2020; but see Rivers et al., 2026). Moreover, although prequestions reliably improve memory for targeted information (King-Shepard et al., 2025; St. Hilaire et al., 2024), some studies also find benefits for information that was not prequestioned (Bostan & Ozcelik, 2026; Rivers et al., 2026; Sana & Carpenter, 2023; St. Hilaire et al., 2019). Any account of the prequestion effect must therefore consider why learning benefits are sometimes restricted to prequestioned information and sometimes extend to other material.

One account that may be able to explain these differences is the *attentional-window account* (Sana & Carpenter, 2023). According to this account, prequestions initiate a period of heightened attention as learners search for the answers, with attention returning to baseline once the relevant information has been encountered. Information presented while this attentional window remains open may also benefit, even if it was not directly targeted by a prequestion. Thus, whether prequestions produce a general benefit should depend partly on where the prequestioned information appears in the learning material. Supporting this prediction, Sana and Carpenter (2023) found that prequestions produced only a specific benefit when the targeted information appeared near the beginning of a passage. However, when the targeted information appeared near the end, prequestions also improved memory for non-prequestioned information encountered while learners were presumably still searching for the answers. These findings suggest that the effects of prequestions may depend on how attention is allocated throughout the learning event. Nevertheless, much of the evidence for this account comes from subsequent memory performance rather than direct observations of processing during learning.

Eye tracking provides a way to examine these attentional predictions as learning unfolds. During reading and visual exploration, the eyes alternate between relatively brief periods of stability, known as fixations, and rapid movements between them, known as saccades. The location and duration of fixations provide measures of overt visual attention and are sensitive to cognitive processes involved in comprehension and learning (Deng & Gao, 2023; Hyönä et al., 2003; Madsen et al., 2021; Rayner, 1998; Rayner, 2009). Because eye movements can be recorded continuously without interrupting the learning experience, eye tracking allows researchers to examine how learners visually process information as they encounter it rather than inferring processing solely from subsequent memory performance (Rayner, 1998). Importantly, eye movements can distinguish processing during an initial encounter with information from subsequent reprocessing, such as rereading a sentence or returning to an earlier part of a passage. Eye-movement measures are also sensitive to individual differences in reading ability (Ashby et al., 2005; Perfetti, 2007), prior knowledge (McNamara & Magliano, 2009), and working memory capacity (Daneman & Carpenter, 1980). Accordingly, real-time measures such as eye tracking have been identified as an important direction for prequestion research (Pan & Carpenter, 2023). In particular, King-Shepard et al. (2025) recommended examining how prequestions influence the time learners spend processing prequestioned and non-prequestioned information during reading.

Relatively few studies have used eye tracking to investigate how prequestions influence reading. In an early study, readers who received questions before reading spent more time rereading and carefully processing question-related information while reading unrelated material more quickly (Rothkopf & Billington, 1979). Lewis and Mensink (2012) examined the effects of prequestions in two eye-tracking experiments using passages on different topics. In Experiment 1, participants read a passage about space travel after receiving one of two sets of six prequestions or no prequestions. Readers receiving one set of prequestions spent more time re-

inspecting and looking back to question-relevant sentences and subsequently recalled more of that information than readers receiving the other set. However, the second set of prequestions did not produce reliable effects on reading or memory. In Experiment 2, Lewis and Mensink (2012) used a passage describing the humanitarian efforts of the Gates Foundation, including efforts to address polio and tuberculosis (TB). One group received four prequestions about TB before reading, whereas a control group read the passage without prequestions. Eye movements were recorded using a head-mounted eye tracker. Prequestions did not increase forward reading times for sentences containing the answers. Instead, readers who received prequestions spent more time rereading the targeted sentences before moving forward and made more look-backs to those sentences after proceeding to later portions of the passage. They also showed a smaller increase in look-backs to other parts of the text. On a subsequent free-recall test, the prequestion group recalled approximately twice as much of the targeted information as the control group, but did not recall more of the non-prequestioned information. Thus, the specific memory benefit was accompanied by greater reprocessing of the targeted sentences rather than slower initial reading. This pattern suggests that readers may first recognize information as relevant to a prequestion and subsequently return to process it more extensively.

More recently, Bostan and Ozcelik (2026) examined the effects of *integrative prequestions*, which require learners to combine information from different parts of a text. Twelve participants received integrative prequestions before reading a passage about car brakes, whereas another twelve read the passage without prequestions. Compared with the control group, readers who received prequestions made more fixations, spent more time on targeted sentences, and made more transitions between them. Unlike most prequestion studies, the authors also observed a general benefit: participants who received prequestions performed better on questions assessing non-prequestioned information despite spending less time fixating those sentences. These findings suggest that prequestions can alter the distribution of visual processing across a

passage, although increased fixation time on particular information does not necessarily correspond to greater memory for that information.

Taken together, previous eye-tracking studies indicate that prequestions influence how learners allocate visual attention during reading (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012; Yang et al., 2021). However, whether these changes contribute directly to improved memory or reflect other processes initiated by prequestions remains uncertain. Furthermore, some findings suggest that the benefits of prequestions cannot be explained simply by the amount of time learners spend processing the material. For example, in a large multisite classroom study, prequestions improved learning even though students subsequently watched less of the instructional videos, with some students skipping the videos entirely (Motz et al., 2026). Similarly, Bostan and Ozcelik (2026) found improved memory for non-prequestioned information despite shorter fixation times on that information. Such findings suggest that prequestions may influence not only the allocation of attention but also how information is processed once encountered.

One alternative explanation is the *mental model hypothesis* proposed by Motz et al. (2026). According to this account, prequestions activate relevant prior knowledge before learning, facilitating the processing and encoding of related information when it is subsequently encountered. As learners recognize connections between the new information and their activated knowledge, they may allocate additional attention to that content. From this perspective, increased attention may follow the recognition of relevant information rather than being the primary mechanism responsible for improved learning. Consistent with this possibility, Motz et al. (2026) found that the benefit of prequestions was greater among students who answered more prequestions correctly, suggesting that access to relevant prior knowledge may contribute to the effect. Although the attention-based and mental model accounts are not necessarily mutually exclusive, they raise an important question about the timing and nature of the processing changes

associated with prequestions. In particular, determining whether prequestions influence reading when relevant information is first encountered or primarily during subsequent reprocessing may help clarify how prequestions shape learning.

The Current Study

The existing literature provides evidence that prequestions influence both memory and the allocation of visual attention during learning. However, questions remain about when these changes in visual processing emerge, whether they are specific to prequestioned information, and how prequestions alter the distribution of reading times. The current study addressed these questions by combining a counterbalanced prequestion design with eye-tracking measures that distinguish initial reading from subsequent reprocessing. We also used distributional analyses to examine how prequestions influence reading times beyond differences in average duration.

Our study builds on two previous eye-tracking investigations of prequestions during reading (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012) while addressing several methodological limitations. First, in Experiment 2 of Lewis and Mensink (2012) and in Bostan and Ozcelik (2026), the same set of sentences was prequestioned for all participants in the prequestion condition. Consequently, the effects of prequestions could not be fully separated from characteristics of the particular sentences being targeted. Although Experiment 1 of Lewis and Mensink (2012) included two sets of prequestions, these targeted different numbers of sentences, and only one set reliably affected reading. In the current study, prequestioned content was fully counterbalanced so that the same sentences were read both with and without prequestions across participants.

Second, previous studies used relatively small samples, with approximately 20 participants per group in Lewis and Mensink (2012) and 12 participants per group in Bostan and Ozcelik (2026), limiting the precision of the estimated effects. In addition, Bostan and Ozcelik (2026) compared total fixation durations and counts between groups without accounting for participant-level differences in reading behavior. The current study included a substantially larger

sample (138 participants, approximately 46 per condition) and used models that accounted for variation across participants and items, including sentences and test questions where appropriate. Third, whereas previous studies primarily analyzed average reading or fixation times, we also examined the distribution of reading times to determine whether prequestions produce broader shifts in reading duration or disproportionately influence unusually long readings.

Fourth, in Lewis and Mensink (2012), participants were presented with questions and instructed to read the passage to answer them, but were not required to provide answers before reading. Although learners may have generated answers covertly, which can be as effective as overtly answering prequestions (Rivers et al., 2026), whether they did so is unknown. In the current study, participants explicitly answered each prequestion before reading, consistent with the procedure used in many prequestion studies. Finally, unlike the integrative questions used by Bostan and Ozcelik (2026), each prequestion in the current study targeted information contained within a single sentence, as in Lewis and Mensink (2012). Such questions have generally been associated with specific rather than general memory benefits (St. Hilaire et al., 2019).

In out study, participants read the Gates Foundation passage used in Experiment 2 of Lewis and Mensink (2012), which describes efforts to eradicate polio and TB. Before reading, participants answered five prequestions about polio, five prequestions about TB, or no prequestions. The passage contained ten target sentences, each providing the answer to one prequestion. By counterbalancing which topic was prequestioned, every target sentence served as prequestioned information for one group and non-prequestioned information for another. This design allowed us to compare processing of the same sentences across prequestion and no-prequestion conditions while minimizing confounding between prequestion status and sentence content.

We examined two categories of eye-movement measures. *Global measures*, including total gaze duration, rereading time, and regressions-in (look-backs), captured processing across encounters with a sentence, including subsequent returns to it. Our *local measure*, selective

regression-path duration, captured processing associated with the initial encounter with a sentence, before the reader progressed to later text. Distinguishing these measures allowed us to assess whether prequestions influence processing when relevant information is first encountered, during subsequent rereading, or both.

In addition to examining these measures, we used distributional analyses to characterize how prequestions influence reading times. Reading and fixation durations are typically right-skewed, with a lower bound of zero and a relatively long tail of unusually long observations (Balota & Yap, 2011; Staub et al., 2010). Consequently, statistical models that assume normally distributed reading times may not adequately characterize the observed data, and transformations intended to reduce skewness can alter the scale on which effects are evaluated (Lo & Andrews, 2015). We therefore modeled total gaze duration using the ex-Gaussian distribution, which represents response times as the sum of a Gaussian component, characterized by its mean (μ), and an exponential component, characterized by its mean (τ).

An advantage of the ex-Gaussian distribution is that it allows changes in the location of the Gaussian component to be distinguished from changes in the exponential component contributing to the right tail. Although these parameters are descriptive and do not uniquely identify underlying cognitive processes (Matzke & Wagenmakers, 2009), previous research suggests that experimental manipulations can influence them differently. For example, word frequency has been found to affect both μ and τ in fixation durations during reading (Staub et al., 2010), whereas modest visual degradation of text primarily affected μ (White & Staub, 2012). Similarly, in scene viewing, image type influenced μ, whereas familiarity increased τ, reflecting a greater contribution from longer fixations (Guy et al., 2020). These findings illustrate how experimental manipulations may alter different characteristics of the reading-time distribution.

This distinction may be particularly informative for understanding prequestion effects. If prequestions lead readers to recognize particular sentences as relevant and sometimes engage in additional processing or reanalysis, they may disproportionately increase longer reading times

rather than uniformly slowing reading. Analyses of average reading duration alone cannot distinguish these possibilities. We therefore examined both ex-Gaussian parameters. An effect on μ would indicate a shift in the Gaussian component of the reading-time distribution, whereas an effect on τ would indicate a greater contribution from the exponential component and its right tail. Although neither pattern provides a unique signature of a cognitive mechanism, these analyses allow a more detailed characterization of how prequestions change reading behavior.

Hypotheses

For final test performance, we predicted a *specific benefit* of prequestions: participants would remember more information about the prequestioned topic than participants in the no-prequestion condition. Based on prior findings that general benefits are less consistent, particularly for questions targeting individual pieces of information (King-Shepard et al., 2025; St. Hilaire et al., 2019), we did not predict a general benefit for the non-prequestioned topic. We compared performance for prequestioned and non-prequestioned information against performance for the same material in the no-prequestion condition, both overall and separately for each topic.

For eye movements, we predicted that prequestioned sentences would receive longer total gaze durations and more rereading than the same sentences encountered without prequestions. Specifically, we expected an interaction between prequestion condition and topic, such that Polio prequestions would selectively increase processing of polio-related sentences, whereas TB prequestions would selectively increase processing of TB-related sentences. We also examined whether these changes emerged during initial reading or subsequent reprocessing. Finally, our distributional analyses assessed whether differences in total gaze duration were associated with changes in the Gaussian component (μ), the exponential component (τ), or both.

Method

We report how we determined our sample size, all data exclusions, all manipulations, and all measures included in the study. The study was preregistered (https://osf.io/8env4). All data,

materials, and analysis scripts are available at https://github.com/jgeller112/Prequestions_EyeTracking. All reported statistics, tables, and figures were generated from the raw data using the code in that repository. Generative AI (Claude Opus 5.5, Anthropic) was used to assist with editing the manuscript and code as well as help organzie all the data and code on the repository. The author verified the content and citations and takes full responsibility for the manuscript.

Participants

The target sample size was determined based on the specific benefit of prequestions. A recent meta-analysis reported a medium-sized effect ($g = 0.54$; St. Hilaire et al. (2024)), which we converted to an odds ratio of 2.66 for the preregistered binomial mixed-effects model. A simulation-based power analysis using mixedpower (Kumle et al., 2021), informed by pilot data, indicated that 45 participants per condition would provide 90% power to detect an effect of this magnitude. We therefore planned to recruit 135 participants, with 45 assigned to each of the three conditions.

Participants were recruited from the Boston College SONA participant pool and received course credit. Eye-tracking data were recorded from 161 participants. One participant had eye-tracking data for fewer than all four pages of the text and was excluded. Eighteen participants did not meet the preregistered calibration criterion (accuracy and precision of 0.5° or better) or could not be calibrated and were excluded. Four participants had no final test data and were excluded. The final sample comprised 138 participants (no prequestions: 47; Polio prequestions: 45; TB prequestions: 46).

Calibration validation data were available for all included participants. Mean calibration accuracy was 0.37° ($SD = 0.10$, range 0.13–0.50), and mean precision, defined as the standard deviation of gaze samples, was 0.22° ($SD = 0.12$, range 0.06–0.49). All included participants with available calibration data met the preregistered criterion of 0.5° or better for both accuracy and precision. Before reading, participants rated their familiarity with polio and tuberculosis on a scale from 0 (*never heard of it*) to 100 (*expert*). Familiarity was low (polio: $M = 24.1$, $SD = 16.8$;

tuberculosis: $M = 29.4$, $SD = 18.8$; ratings were available for 109 of the 138 participants[1]) and similar across conditions, so the preregistered exclusion based on prior knowledge, for which no criterion had been specified, was not applied.

Materials

Participants read a 690-word passage used by Lewis and Mensink (2012) describing the Gates Foundation's humanitarian efforts to eradicate polio and tuberculosis (TB) in developing countries. The passage contained 38 sentences presented across four pages, each consisting of 16–17 lines (Figure 1). Ten sentences served as targets: five concerning polio ($M = 19.4$ words, $SD = 11.4$, range 4–35) and five concerning TB ($M = 16.2$ words, $SD = 3.7$, range 11–21).

The polio target sentences had a higher reading level than the TB target sentences, as assessed by the Flesch–Kincaid grade level (Kincaid et al., 1975), calculated using the quanteda package (Benoit et al., 2018) (polio: $M = 14.1$, $SD = 6.2$; TB: $M = 8.8$, $SD = 2.0$). However, the two sets of sentences were similar in syntactic complexity, estimated using dependency parsing with UDPipe (Straka & Straková, 2017; Wijffels & Straka, 2026) and its English Web Treebank model (Silveira et al., 2014). We examined mean dependency distance (Liu, 2008), defined as the average number of words separating a word from its syntactic head (polio: $M = 2.72$, $SD = 1.03$; TB: $M = 2.62$, $SD = 0.40$), and the number of clauses per sentence (polio: $M = 1.6$, $SD = 0.9$; TB: $M = 1.8$, $SD = 0.8$).

Each target sentence contained the answer to one of ten short-answer questions (five about polio and five about TB; Appendix A). Questions required percentage-based, yes/no, or multipart responses. The maximum possible score was 12 points per topic and 24 points overall.

The study included three conditions: polio prequestions, TB prequestions, and no prequestions. Participants in the polio and TB conditions answered five questions about their assigned topic before reading, whereas participants in the no-prequestion condition read the

[1] Survey responses were linked to the eye-tracking data by participant name. Ratings are missing for participants whose name was not recorded on the session log or who did not enter their name in the survey.

passage without answering questions beforehand. For each prequestion condition, the five target sentences corresponding to the assigned topic were classified as *prequestioned*, and the five sentences about the other topic were classified as *non-prequestioned*. Thus, across conditions, every target sentence served as both prequestioned and non-prequestioned information, allowing the effects of prequestions to be separated from differences in sentence content.

Apparatus

Eye movements were recorded using a Tobii Pro Spectrum eye tracker sampling at 600 Hz. The passage was displayed on a 23.8-inch EIZO FlexScan EV2451 monitor (1920 × 1080 pixels) viewed from approximately 65 cm without a chin rest. Text was presented in 36-point Courier New, a monospaced font, against a grey background. At this viewing distance, each character subtended approximately 0.5° of visual angle, and the line spacing subtended approximately 1.1°.

The experiment was administered using Tobii Pro Lab (version 25.19). The prequestions, distractor task, and final test were programmed in PsychoPy, hosted on Pavlovia, and embedded as web elements within the Tobii Pro Lab timeline.

Procedure

At the beginning of the session, participants were randomly assigned to one of three conditions: polio prequestions, TB prequestions, or no prequestions. The experiment consisted of four phases: prequestions, reading, a distractor task, and a final test. Participants in the no-prequestion condition skipped the prequestion phase.

Prequestions

Participants in the two prequestion conditions answered five short-answer questions about their assigned topic (polio or TB), presented in random order (https://run.pavlovia.org/Jgeller112/preq). They were informed that they might not know the correct answers and were encouraged to provide their best guesses. Each question appeared individually on the screen, with a text box below it for entering a response. Participants had unlimited time to respond and

clicked a triangle at the bottom of the screen to advance to the next question. No feedback was provided.

Reading

Before reading, participants completed Tobii Pro Lab's nine-point calibration procedure, which presented white targets against a grey background. Calibration accuracy and precision were then validated and recorded. Participants subsequently read the four-page passage while their eye movements were recorded. They advanced through the pages at their own pace, with no time limit, but could not return to previously viewed pages.

Distractor Task

After reading, all participants completed a president recognition task (https://run.pavlovia.org/Jgeller112/preq_president) that served as a distractor and lasted approximately 3 minutes.

Final Test

Participants completed a final test consisting of all ten short-answer questions (five about polio and five about TB), presented in random order (https://run.pavlovia.org/Jgeller112/preQ_test). The response format was identical to that used during the prequestion phase, with unlimited response time and no feedback. For participants in the prequestion conditions, five questions were repeated from the prequestion phase, whereas the remaining five were new. Participants were then thanked and debriefed.

Scoring

Prequestion and final-test responses were scored using a deterministic, rule-based scoring procedure developed with the assistance of Claude Opus 5.5 (Anthropic; model ID `claude-opus-5-5`). Rather than scoring individual responses directly, Claude generated an R scoring script (`scoring_rubric.R`, used by `grade_prequestions.R` and `grade_test.R`) that applied the same predefined criteria to all responses. The script awarded one point for each required element

of a correct answer present in a participant's response. Question-specific scoring criteria are provided in Appendix A.

To assess scoring reliability, a research assistant who was blind to experimental condition, test phase, and automated scores independently scored all 1,678 nonblank responses. The automated and human scores agreed exactly on 97.6% of responses. Agreement was excellent, ICC(2,1) = .99, 95% CI [.99, .99], based on a two-way random-effects, absolute-agreement, single-measure intraclass correlation coefficient (Koo & Li, 2016; Shrout & Fleiss, 1979), calculated using the irr package (Gamer et al., 2026). Agreement for participants' total final-test scores was also excellent, ICC(2,1) = .99. Question-level agreement statistics are reported in Table A2. All analyses used the automated scores.

Eye-Movement Data Processing

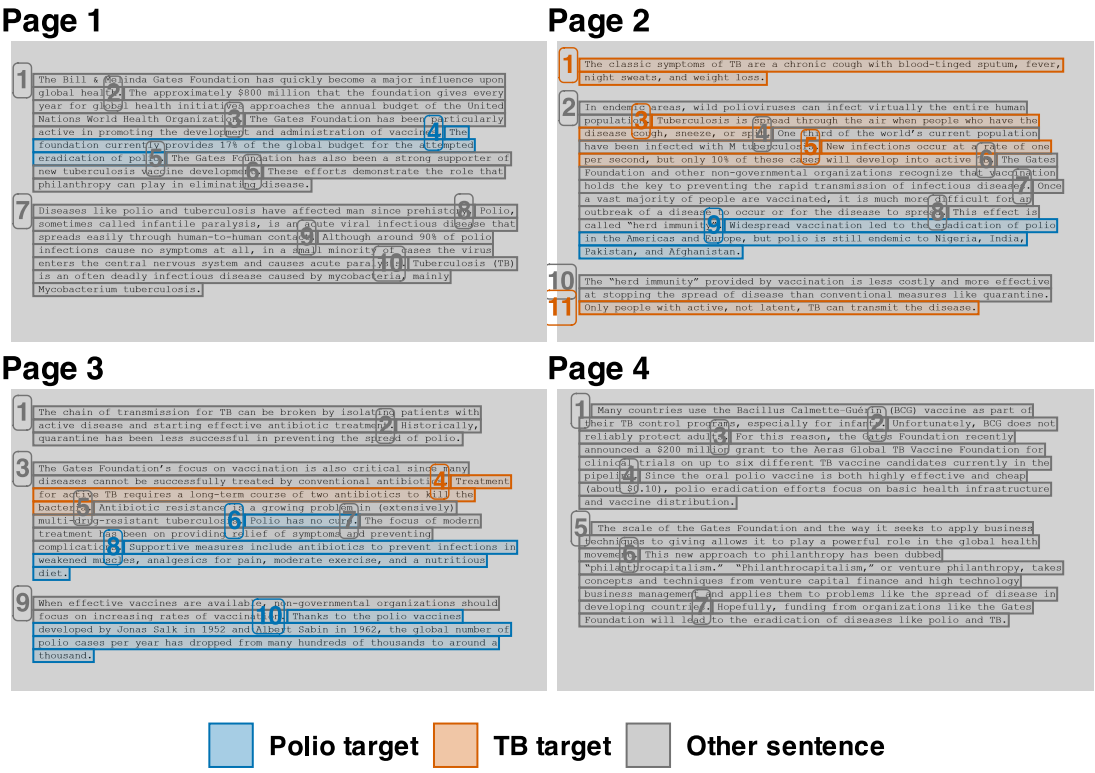
Sentence-level measures were computed from the fixation-level data. Fixations were identified with Tobii Pro Lab's Tobii I-VT (Fixation) filter, a velocity-threshold identification algorithm (Olsen, 2012; Salvucci & Goldberg, 2000), with the following settings. Gaps in the gaze data shorter than 75 ms (e.g., brief losses of tracking) were filled by linear interpolation, the gaze positions of the two eyes were averaged, and noise was reduced with a moving median over a window of three samples. A gaze sample was classified as part of a fixation when the angular velocity of the eye, computed over a 20-ms window, was below 30°/s. Adjacent fixations separated by less than 75 ms and less than 0.5° of visual angle were merged, and fixations shorter than 60 ms were discarded. Fixations cut off when a page appeared or disappeared can nonetheless be shorter than 60 ms in the export. We excluded fixations shorter than 80 ms (4.0% of fixations) and longer than 1,000 ms (0.06%) before any further processing. Both bounds follow common practice in reading research: 80 ms is the most common lower bound, and 800, 1,000, and 1,200 ms are the most common upper bounds (Schotter & Dillon, 2025; e.g., Serrano-Carot et al., 2025). Very short fixations are unlikely to reflect processing of the text and include

fixations cut short by a page change. Most fixations in reading last around 200 ms (Rayner, 1998; Schotter & Dillon, 2025); in the present data the median fixation lasted 178 ms, so a single fixation longer than 1,000 ms, 5.6 times the median, is unlikely to reflect ongoing reading and more likely reflects a lapse of attention or a brief loss of the eye-tracking signal that the fixation filter bridged.

The remaining processing (`derive_et_measures.R`) proceeded in four steps. (a) The exact layout of each page was reconstructed from the sentence texts and full-resolution screenshots of the pages (monospaced font, 79 characters per line), giving the screen position of every character of every sentence (Figure 1). (b) Each fixation was assigned to a text line after correcting for vertical drift, separately for each reader and page, with a procedure that combines two approaches used in automated drift-correction algorithms (Carr et al., 2022): assigning sequences of fixations to a line together, and modelling drift as a function of position on the page (details in Appendix D). The correction was limited to the calibration accuracy criterion of 0.5° (0.57 cm, about 0.45 of a line spacing), so it could not move a reader's gaze by a whole line. (c) Each fixation was then assigned to the sentence occupying its line and horizontal position: its character column was computed from the fixed character width, a fixation on a space took the sentence of the nearest letter, and fixations more than three characters beyond either end of a line were treated as off the text. (d) Reading on each page was taken to start at the first fixation on the page's first sentence; fixations before that (the landing) contributed to total time but not to first-pass measures. After processing, 96.8% of on-screen fixations were assigned to a sentence, and the assigned sentence agreed with Tobii Pro Lab's assignment for 89.8% of fixations. First-pass reading was identified for 96.9% of fixated target sentences. Figure 2 shows one reader's fixations on one page before and after processing. For this reader, the correction moved fixations by a median of 0.44 line spacings and changed the sentence assigned to 43.4% of fixations; the medians over all readers and pages were 0.24 line spacings and 8.2%.

Figure 1

The Four Text Pages and the Sentence Regions Used as Areas of Interest



Note. Each sentence is one area of interest, made of one rectangle per text line it occupies. Polio target sentences are shown in blue, TB target sentences in vermillion (orange-red), and other sentences in grey; numbers give each sentence's position on its page. Page 4 contains no target sentences.

Figure 2

One Reader's Fixations on One Page Before and After Processing



Note. One participant in the TB prequestion condition, page 2, chosen to illustrate vertical drift. (A) Recorded gaze positions, coloured by the sentence assigned by Tobii Pro Lab. (B) The same fixations after drift correction (single vertical offset, assignment of chains of fixations to lines,

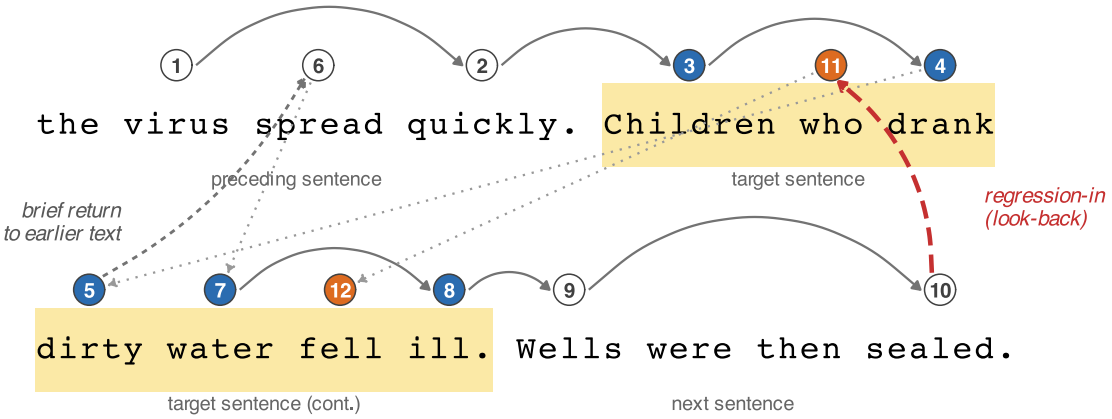
and a linear drift model, limited to 0.5°; see Appendix D), coloured by the sentence assigned in processing. Each point is one fixation (size proportional to duration); grey points were not assigned to any sentence. Lines join consecutive fixations. Open circles are fixations made before reading started on the page. Dashed rectangles outline the target sentences.

Eye-Movement Measures

All measures were computed for each participant and target sentence. Durations were divided by the number of words in the sentence (ms per word), because target sentences differed in length. Only sentences that were fixated were analysed: a target sentence that received no fixation has no reading time, so its measures were treated as missing (17 of 1380 target sentences, 1.2%). Following the eye-movement literature, we distinguish *global* measures, which reflect all processing of a sentence including later returns to it, from *local* measures, which reflect processing when the sentence is first encountered (Figure 3).

Figure 3

Eye-Movement Measures, Illustrated With a Schematic Reading Example



Measure	Type	Fixations in the example	Analysed as
Selective regression-path duration	Local	3 + 4 + 5 + 7 + 8	Ex-Gaussian (μ, τ)
Total gaze duration	Global	3 + 4 + 5 + 7 + 8 + 11 + 12	Ex-Gaussian (μ, τ)
Regression-in (look-back)	Global	1 (move from 10 to 11)	Logistic
Re-reading time	Global	11 + 12	Gamma (if re-read)
Fixation on another sentence		not counted for the target	

Note. Definitions follow Schotter and Dillon (2025). The circles are fixations, numbered in time order (arrows: saccades; dashed arrows: regressions; dotted arrows: moves to the next line); the shaded sentence is the target, and each fixation is coloured by the measure it counts towards for the target. *Selective regression-path duration* (first-pass reading; also called right-bounded reading time or selective go-past time) sums the fixations on the sentence from first entering it until the eyes first move past it to later text; fixations on earlier text during this period (here, 6) are not counted and do not end it. *Total gaze duration* sums all fixations on the sentence. *Regression-in* records whether the eyes returned to the sentence from later text (0/1). *Re-reading time* sums the fixations on the sentence after the reader had first moved on to later text (operationalizing the preregistered "re-reading durations back to the prequestioned information"). Selective regression-path duration is defined only for sentences fixated before any

later sentence (not skipped). The duration measures do not overlap: total gaze duration is the sum of selective regression-path duration, re-reading time, and any fixations made before reading started on the page. Global measures include later returns to the sentence; the local measure reflects processing when the sentence is first encountered. The preregistered first-pass duration (fixations until the sentence is first exited) would here be 3 + 4 + 5 only, cut short by the brief return at 6; it was replaced by selective regression-path duration (see Statistical Analysis). All durations were divided by the number of words in the sentence (ms per word).

Statistical Analysis

All analyses were conducted in R version 4.6.1 (R Core Team, 2026).[2] The effects of prequestions on test performance and eye movements were estimated with Bayesian mixed models in brms (Bürkner, 2017), which fits models in Stan (B. Carpenter et al., 2017), run through cmdstanr (Gabry et al., 2025) with CmdStan 2.37.0. Each target sentence, and each final-test question, was coded by its status for the reader: *prequestioned* (Polio sentences and questions for readers in the Polio condition, TB sentences and questions for readers in the TB condition), *non-prequestioned* (the other topic for those readers), or *no-prequestion* (both topics for readers in the no-prequestion condition). The main models pooled the two prequestion conditions, with Status and Topic (Polio, TB) as fixed effects. The *specific effect* of prequestions was the difference prequestioned – no-prequestion: the same sentences or questions for readers whose prequestions they answered and for readers who saw no prequestions. The *general effect* was non-prequestioned – no-prequestion. Because each topic contributes equally to the prequestioned, non-prequestioned, and no-prequestion cells, differences between the Polio and

[2] Data were processed and analysed in R version 4.6.1 (R Core Team, 2026) with the packages tidyverse 2.0.0 (Wickham et al., 2019), janitor 2.2.1 (Firke, 2024), png 0.1.9 (Urbanek, 2026), modelbased 0.16.0 (Makowski et al., 2025), marginaleffects 0.32.0 (Arel-Bundock et al., 2024), insight 1.5.2 (Lüdecke et al., 2019), brms 2.23.0 (Bürkner, 2017), cmdstanr 0.9.0 (Gabry et al., 2025), cogmod 0.3.3 (Makowski, 2026), bayestestR 0.18.1 (Makowski, Ben-Shachar, & Lüdecke, 2019), priorsense 1.4.0 (Kallioinen et al., 2023), ggdist 3.3.3 (Kay, 2024), quanteda 4.5.0 (Benoit et al., 2018), quanteda.textstats 0.97.2 (Benoit et al., 2024), udpipe 0.8.16 (Wijffels & Straka, 2026), irr 0.85 (Gamer et al., 2026), patchwork 1.3.2 (Pedersen, 2025), scales 1.4.0 (Wickham et al., 2025), knitr 1.51 (Xie, 2015), and xfun 0.60 (Xie, 2026). The manuscript was produced with Quarto 1.10.18 and the apaquarto extension.

TB materials cancel in these comparisons. Both the specific and general effects and their breakdown by topic were preregistered. To break the effects down by topic, we fitted the preregistered Group (Polio prequestions, TB prequestions, or no prequestions) × Topic (Polio, TB) models with the same families and priors. Following the preregistered hypotheses, we estimated the Polio – TB difference within each group—predicted to be positive in the Polio group, negative in the TB group, and absent in the no-prequestion group—and the Group × Topic interaction contrast, the Polio group's Polio – TB difference minus the TB group's, predicted to be positive. Because the Polio and TB sentences (and questions) are different items, each within-group Polio – TB difference combines any effect of prequestions with differences between the items, and with five items per topic, the population-level topic difference is estimated imprecisely. Each within-group difference was therefore interpreted together with the interaction contrast, in which the item differences cancel because both groups read the same items.

Reading times are strongly right-skewed, so no duration was analysed with a Gaussian model. As preregistered for heavily skewed durations, total gaze duration was analysed with an ex-Gaussian distributional model ({cogmod}; Makowski (2026)), which describes a duration as the sum of a Gaussian component, whose mean μ reflects typical reading time, and an exponential component, whose mean τ reflects the tail of unusually long readings. The mean of an ex-Gaussian distribution is $\mu + \tau$ and its standard deviation is $\sqrt{(\sigma^2 + \tau^2)}$, where σ is the standard deviation of the Gaussian component, so μ and τ are components of reading time rather than reading times themselves. The fixed and random effects applied to both μ and τ, and σ had by-participant random intercepts. The preregistered first-pass duration, from first entering a sentence until first leaving it, could not be described by any of the distributions we considered (ex-Gaussian, Gamma, lognormal), because it mixes two kinds of first pass. About half of first passes, in all three conditions, were cut short after a few fixations by a brief return to the preceding line, so the distribution had two modes: interrupted first passes of a few hundred

milliseconds and complete first readings of several seconds. First-pass reading was therefore measured as selective regression-path duration [also called right-bounded reading time; Schotter and Dillon (2025)]: the time spent on the sentence from first entering it until the eyes first moved past it to later text, so that brief returns to earlier text did not end it and time spent on earlier text was not counted (Figure 3). Its distribution was unimodal and right-skewed and closely described by the ex-Gaussian, so it was analysed with the same ex-Gaussian model as total gaze duration (μ and τ). Re-reading time was the time spent on a sentence after the reader had first moved on to later text, so that the two measures do not overlap (cf. rereading time in Schotter & Dillon, 2025). Whether readers returned to a sentence from later text is captured by regressions-in (0/1), which were analysed with logistic models; re-reading time was analysed for the 37.5% of sentences that were re-read, with a Gamma model with a log link, which models positive, right-skewed durations without transforming them (Lo & Andrews, 2015); the ex-Gaussian was not suitable, because re-reading times were close to exponentially distributed, so that its Gaussian component collapsed (σ near zero). The model therefore describes how long readers re-read a sentence once they had returned to it. Because the questions were worth different numbers of points (1–4), final test performance was analysed as the points earned on each question out of its maximum, with a beta-binomial model: a binomial model of the points earned with an additional parameter for overdispersion, because a participant tends to know all or none of a question's elements. Effects are reported on the scale of the expected proportion of a question's points, which puts all questions on the same 0–1 scale. Because each question has at most four points, scores can take only a few values (0, .25, .33, .50, .67, .75, and 1). In posterior predictive checks, the beta-binomial model reproduced how often each score occurred (predicted and observed shares within 1.1 percentage points), better than a binomial model without overdispersion. Prequestion performance was analysed in the same way, comparing the two prequestion conditions.

All models included the maximal random-effects structure warranted by the design (Barr et al., 2013). Topic varied within participants, so the models included by-participant random intercepts and Topic slopes (the preregistered structure). Every target sentence and every final-test question was prequestioned for one group, non-prequestioned for another, and read or answered by the no-prequestion group, so Status (and Group) varied within items, and the models also included by-sentence (eye movements) or by-question (final test) random intercepts and Status (or Group) slopes. Within a participant, Status is determined by Topic, so no by-participant Status slope was possible. In the prequestion model, condition varied between participants and between questions, so it included random intercepts for participants and questions only. Status, Condition, and Topic were coded with equal-prior contrasts [`contr.equalprior_pairs`; {bayestestR}, Makowski, Ben-Shachar, and Lüdecke (2019)]. With this coding, a normal(0, s) prior on the coefficients implies the same normal(0, s) prior on every pairwise difference between conditions, so that no condition serves as a reference with a different prior, and every comparison is tested against the same prior. s was set on each model's link scale: 1.5 on the log-odds scale (regressions-in, final test, and prequestions; an odds ratio of 4.5 for one *SD*), 0.5 on the log scale of the Gamma models and on the scale of τ (a factor of about 1.65 for one *SD*), and 100 ms per word for μ. Intercepts, random-effect standard deviations, and correlations had weakly informative priors (Appendix B). Each model was sampled with four chains of 10000 post-warm-up iterations (40000 posterior draws, enough for stable Bayes factors). All models converged: there were no divergent transitions, all R-hat values were below 1.001, and all bulk effective sample sizes exceeded 4900.

Means and comparisons were computed on the response scale by averaging each posterior draw's predictions over the observed sample (modelbased with the {marginaleffects} (Arel-Bundock et al., 2024) backend and `estimate = "average"`), so that they incorporate the variation due to the random effects. Durations are reported in ms per word, and probabilities and

proportions in percent. Effects were summarized by their posterior mean, credible interval (CrI), and probability of direction (pd), the proportion of the posterior with the same sign as its median. The pd is closely related to the frequentist p value: for a one-tailed test, $p \approx 1 - \text{pd}$, and for a two-tailed test, $p \approx 2(1 - \text{pd})$, so pd > 95% corresponds to a one-tailed $p < .05$ (Makowski, Ben-Shachar, Chen, et al., 2019). The predictions for the specific effects and for prequestioned versus non-prequestioned material were directional (longer reading, more regressions, better recall), so we treated such an effect as present when pd > 95% in the predicted direction and report it with a 90% CrI, the interval that corresponds to a one-sided test at $\alpha = .05$. The general effects had no directional prediction, so for them we required pd > 97.5%, corresponding to a two-tailed $p < .05$, and report a 95% CrI. Comparisons that were described but not tested are also reported with a 95% CrI. Bayes factors were not used as a decision criterion; we report them to describe the strength of the evidence for or against each effect. Because pd only indicates the direction of an effect and cannot provide evidence that an effect is absent, we also computed Bayes factors. Each Bayes factor compares a model in which the comparison equals 0 with one in which it is free, using the Savage–Dickey density ratio (Wagenmakers et al., 2010): the posterior density of the comparison at 0 divided by its prior density at 0. The prior densities came from the same models sampled from their priors only, and both densities were computed for the comparisons on the link scale, where the equal priors apply. For τ and for the expected re-reading time, which have no single link scale, they were computed on the response scale; for the expected re-reading time, at the population level (without random effects), because averaging over random effects drawn from their priors produced prior draws too extreme for a density estimate. For the Group × Topic comparisons of look-backs and final test scores, the Bayes factors were also computed on the response scale (probabilities and proportions of points), the scale of the estimates. These comparisons involve different Polio and TB items, some of which had outcomes near 0 or 1, and for such outcomes the same comparison is much less precise in log-odds than in probability, so

link-scale Bayes factors could contradict the estimates. $BF_{10} > 3$ indicates moderate, $BF_{10} > 10$ strong, $BF_{10} > 30$ very strong, and $BF_{10} > 100$ extreme evidence for a difference, and $BF_{01} = 1/BF_{10}$ the corresponding evidence for no difference; values between 1/3 and 3 are anecdotal (Lee & Wagenmakers, 2014). Because a Bayes factor depends on the prior, all comparisons between conditions received the same prior, and we report how sensitive the posteriors were to the priors [power-scaling sensitivity; Kallioinen et al. (2023)].

Deviations From the Preregistration

Deviations from the preregistration, the reasons for them, and their likely effect on the results are listed in Appendix C (Table C1).

Results

Prequestion Performance

As intended, participants knew little of the material before reading. Participants in the Polio prequestion condition earned 4.1% of the available points on the prequestions (*SD* = 5.2%) and those in the TB prequestion condition 10.5% (*SD* = 10.9%). Participants attempted 43.1% (Polio) and 39.6% (TB) of the prequestions; the rest were left blank or answered "I don't know". In a beta-binomial model of the points earned on each prequestion, with random intercepts for participants and questions, prequestion scores were similar in the two conditions (model-estimated scores: 10.6% in the TB and 8.6% in the Polio condition), TB – Polio: 2.0 percentage points, 95% CrI [–2.0, 6.0], pd = 83.0%.

Final Test Performance

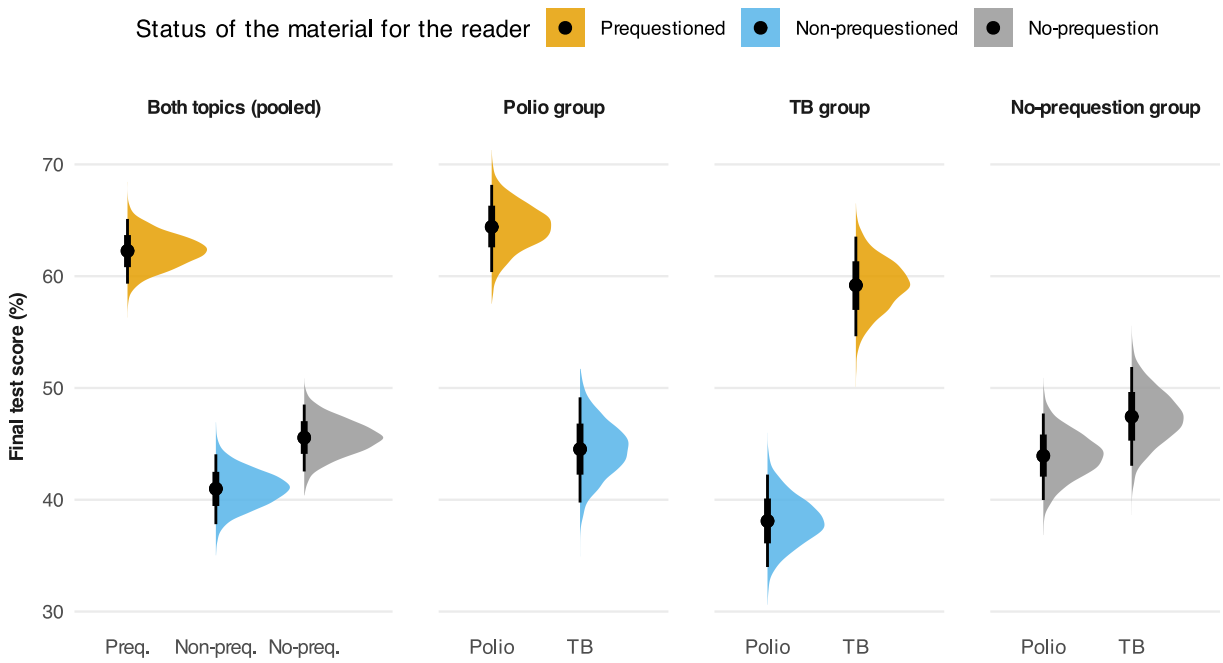
Final test performance was analysed as the proportion of each question's points earned (Figure 4). Pooled over the two prequestion conditions, participants earned 62.2% of the points on their prequestioned topic, compared with 45.6% for no-prequestion participants on the same questions, a specific benefit of 16.7 percentage points, 90% CrI [13.2, 20.1], pd > 99.9%, $BF_{10} = 1.3 \times 10^{8}$: extreme evidence for a difference. On the non-prequestioned topic they earned 41.0%, numerically slightly lower than in the no-prequestion condition, –4.6 percentage points, 95% CrI

[−8.9, −0.3], pd = 98.1%, BF_{01} = 1.85. There was therefore no general benefit. Although the difference met the preregistered criterion (pd > 97.5%) in the direction of a cost, the Bayes factor was inconclusive, providing only anecdotal evidence for no difference, so we do not interpret it as a general cost.

In the Group × Topic model (Figure 4; Bayes factors on the response scale, see Statistical Analysis), the Polio group scored higher on Polio than on TB questions (64.4% vs. 44.5%), Polio − TB = 19.9 percentage points, 95% CrI [13.8, 26.0], pd > 99.9%, $BF_{10} = 8.6 \times 10^{5}$, whereas the TB group scored higher on TB than on Polio questions (59.2% vs. 38.1%), Polio − TB = −21.1 percentage points, 95% CrI [−27.1, −15.1], pd > 99.9%, $BF_{10} = 1.6 \times 10^{6}$. The no-prequestion group, which saw no prequestions, scored 43.9% on Polio and 47.5% on TB questions, Polio − TB = −3.6 percentage points, 95% CrI [−9.5, 2.3], pd = 88.3%, BF_{01} = 3.54. The Group × Topic interaction contrast—the Polio group's Polio − TB difference minus the TB group's—was 41.0 percentage points, 90% CrI [33.8, 48.2], pd > 99.9%, $BF_{10} = 1.5 \times 10^{10}$: extreme evidence that each prequestion group did relatively better on its own topic.

Figure 4

Final Test Performance: Posterior Distributions of the Model-Estimated Means



Note. Posterior distributions of the model-estimated proportion of a question's points earned, averaged over the observed sample. Left: the status model, pooled over the two prequestion groups and both topics (prequestioned, non-prequestioned, and no-prequestion). Other panels: the Group × Topic model, for the Polio and TB questions in the Polio, TB, and no-prequestion groups; colours give the status of the questions for the readers in that group. Preq. = prequestioned; non-preq. = non-prequestioned; no-preq. = no-prequestion. Points are posterior means; thick and thin bars are 66% and 95% credible intervals.

Eye Movements

Without prequestions, readers spent on average 249 ms per word on a target sentence in total (*SD* = 136; about 241 words per minute), 215 ms per word (*SD* = 109) of it in first-pass reading (selective regression-path duration). The ex-Gaussian estimates below divide such reading times into a typical part (μ) and a tail (τ), which sum to the mean. We first compare prequestioned and non-prequestioned sentences with the same sentences read by no-prequestion

readers (Table 1), and then test the preregistered Group × Topic crossover (Figure 5). Figure 5 shows the posterior distributions of the means for every measure, pooled and by group.

Prequestioned and Non-Prequestioned Sentences Versus the No-Prequestion Condition

Prequestions changed how readers read the sentences that answered them, compared with no-prequestion readers reading the same sentences (Table 1). During first-pass reading, measured as selective regression-path duration, typical reading time did not differ, μ: −5.0 ms per word, 90% CrI [−17.2, 7.2], pd = 75.0%, BF_{01} = 12.13, with strong evidence for no difference. The tail of unusually long first-pass readings, however, was longer for prequestioned sentences, τ: 17.1 ms per word, 90% CrI [1.5, 32.8], pd = 96.4%, BF_{10} = 1.25, although the Bayes factor provided only anecdotal evidence for a difference. Total gaze duration showed the same pattern more clearly: prequestioned sentences had a longer tail, τ: 33.9 ms per word, 90% CrI [15.4, 52.7], pd = 99.9%, BF_{10} = 31.07, very strong evidence for a difference, whereas typical reading time did not differ, μ: 2.8 ms per word, 90% CrI [−9.2, 14.8], pd = 64.7%, BF_{01} = 14.26, with strong evidence for no difference. After moving on, readers were more likely to return to prequestioned sentences from later text (37.6% vs. 30.2%), 7.3 percentage points, 90% CrI [3.2, 11.4], pd = 99.8%, BF_{10} = 1.67, anecdotal evidence, and when they returned, they re-read prequestioned sentences for longer (129.2 ms per word vs. 100.0 ms per word), 29.2 ms per word, 90% CrI [11.2, 47.6], pd = 99.6%, BF_{10} = 9.49, moderate evidence.

Non-prequestioned sentences, by contrast, were read like the same sentences in the no-prequestion condition. No measure showed a general effect (non-prequestioned − no-prequestion, Table 1; all pd < 97.5%), and on every measure the Bayes factor favoured its absence (BF_{01} from 3.7 to 15.4), with moderate or stronger evidence on every measure. The effects of prequestions on reading were therefore specific to the prequestioned sentences.

Table 1

Prequestioned and Non-Prequestioned Sentences Versus the Same Sentences Read Without Prequestions

Measure	Prequestioned – No-prequestion	pd	BF_{10}	Non-prequestioned – No-prequestion	pd	BF_{01}
Selective regression path: μ	–5.0 [–17.2, 7.2]	75.0%	0.08	–4.9 [–19.0, 9.2]	75.5%	12.27
Selective regression path: τ	17.1 [1.5, 32.8]	96.4%	1.25	–0.1 [–17.1, 17.1]	50.3%	4.62
Total gaze: μ	2.8 [–9.2, 14.8]	64.7%	0.07	2.3 [–11.2, 15.9]	63.1%	15.41
Total gaze: τ	33.9 [15.4, 52.7]	99.9%	31.07	–4.7 [–24.0, 14.7]	68.5%	3.72
Regression-in	7.3 [3.2, 11.4]	99.8%	1.67	–1.3 [–6.1, 3.5]	70.6%	6.12
Re-reading time	29.2 [11.2, 47.6]	99.6%	9.49	0.3 [–19.8, 20.6]	51.2%	5.56

Note. Posterior means and credible intervals of the differences (90% for the directional specific effect, prequestioned – no-prequestion; 95% for the nondirectional general effect, non-prequestioned – no-prequestion), pooled over the Polio and TB prequestion conditions and averaged over the two topics, in ms per word (durations) or percentage points (regressions-in). Prequestioned – no-prequestion is the specific effect and non-prequestioned – no-prequestion the general effect of prequestions. Selective regression-path duration (first-pass reading) and total gaze duration were analysed with ex-Gaussian models (μ = typical reading time; τ = tail of unusually long readings); regressions-in with a logistic model; re-reading time, for sentences that were re-read, with a Gamma model. pd = probability of direction. BF_{10} = Bayes factor for a difference (Savage–Dickey); $BF_{01} = 1/BF_{10}$, evidence for no difference. Selective regression-path duration is a local measure; all others are global.

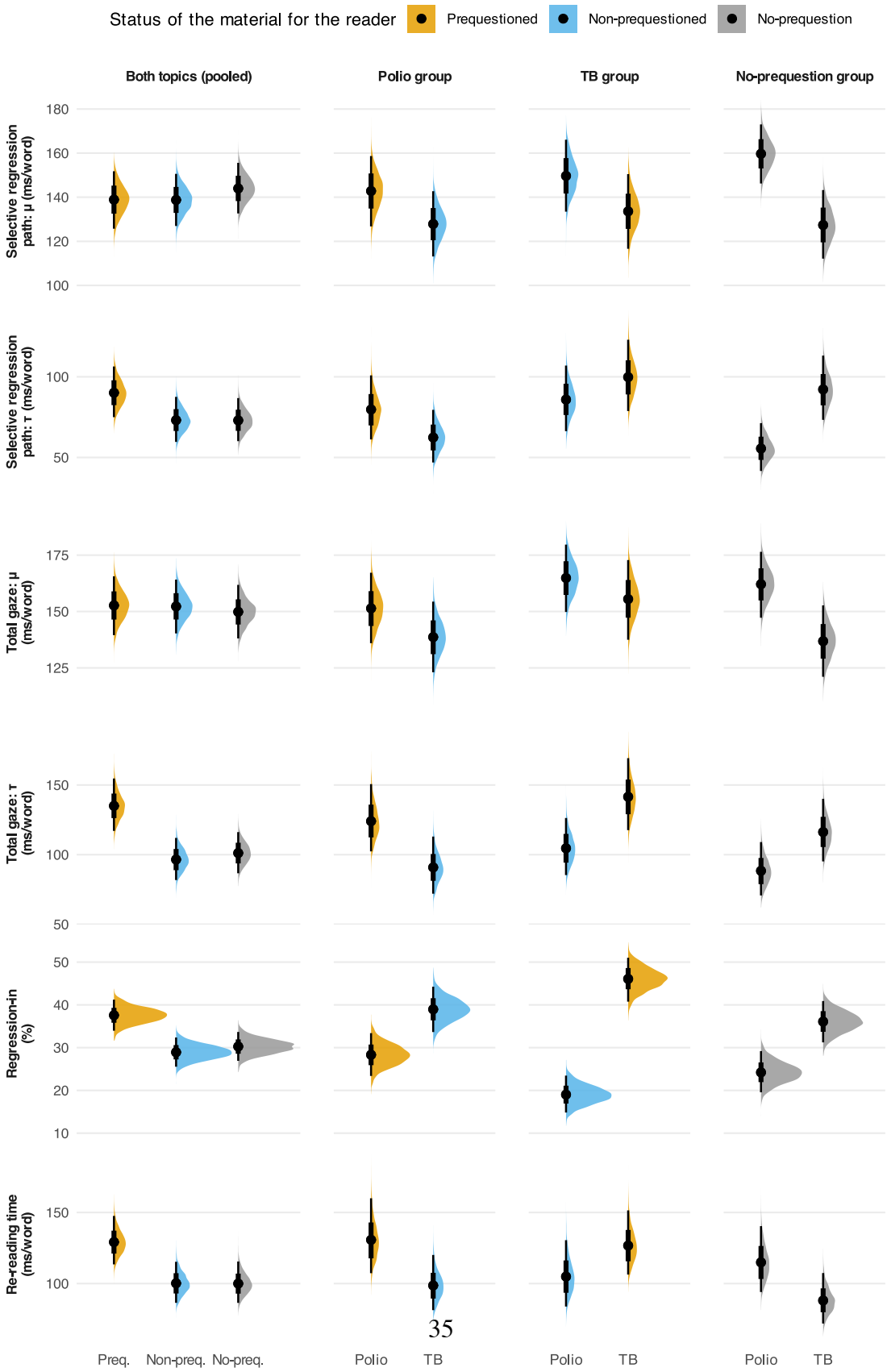
Group × Topic

The preregistered prediction was a crossover: longer reading of Polio than of TB sentences in the Polio group, the reverse in the TB group, and no difference in the no-prequestion group, so that the Group × Topic interaction contrast would be positive. The interaction contrast was positive (pd > 95%) for the tail of first-pass reading (τ): 31.1 ms per word, 90% CrI [2.0, 60.7], pd = 96.1%, BF_{10} = 1.68; the tail of total reading time (τ): 70.1 ms per word, 90% CrI [35.6, 105.2], pd = 99.9%, BF_{10} = 109; look-backs from later text: 16.5 percentage points, 90% CrI [8.4, 24.5], pd > 99.9%, BF_{10} = 63.06; and re-reading time: 54.0 ms per word, 90% CrI [17.8, 90.2], pd = 99.3%, BF_{10} = 4.79. It was not reliably positive for typical first-pass reading time (μ): −1.1 ms per word, 90% CrI [−23.4, 21.2], pd = 53.5%, BF_{01} = 9.40 and typical total reading time (μ): 3.5 ms per word, 90% CrI [−19.0, 26.1], pd = 60.1%, BF_{01} = 9.06. For the tail of total gaze duration (τ), the Polio − TB difference was 33.2 ms per word, 95% CrI [4.6, 62.7], pd = 98.9%, BF_{10} = 4.24 in the Polio group, −36.9 ms per word, 95% CrI [−68.6, −6.1], pd = 99.0%, BF_{10} = 5.07 in the TB group, and −27.9 ms per word, 95% CrI [−55.9, −1.0], pd = 97.9%, BF_{10} = 2.27 in the no-prequestion group. For look-backs from later text (Bayes factors on the response scale), it was −10.6 percentage points, 95% CrI [−17.7, −3.4], pd = 99.8%, BF_{10} = 10.73 in the Polio group, −27.2 percentage points, 95% CrI [−33.7, −20.5], pd > 99.9%, $BF_{10} = 2.2 \times 10^{7}$ in the TB group, and −11.9 percentage points, 95% CrI [−18.6, −5.1], pd > 99.9%, BF_{10} = 58.69 in the no-prequestion group. For re-reading time, it was 32.2 ms per word, 95% CrI [1.5, 64.5], pd = 98.0%, BF_{10} = 2.99 in the Polio group, −21.8 ms per word, 95% CrI [−54.1, 11.3], pd = 90.7%, BF_{01} = 3.64 in the TB group, and 26.6 ms per word, 95% CrI [−1.5, 56.6], pd = 96.8%, BF_{10} = 2.51 in the no-prequestion group. Because the Polio and TB sentences differ, the no-prequestion group shows how the topics are read without prequestions (Figure 5). Compared with that group, the crossover in look-backs came mainly from the TB group, who looked back to TB sentences more often (46.1% vs. 36.1%) and to Polio sentences less often (19.0% vs. 24.2%) than no-

prequestion readers, whereas the Polio group's look-backs differed little from those of no-prequestion readers on either topic (Polio: 28.4% vs. 24.2%; TB: 39.0% vs. 36.1%). Re-reading time showed the same asymmetry: the TB group re-read TB sentences for longer than no-prequestion readers (126.9 ms per word vs. 88.1 ms per word), whereas the Polio group's re-reading of Polio sentences was closer to that of no-prequestion readers (130.8 ms per word vs. 114.7 ms per word). These comparisons across groups are descriptive.

Figure 5

Eye-Movement Measures: Posterior Distributions of the Model-Estimated Means



Note. Posterior distributions of the model-estimated means, averaged over the observed sample. For the ex-Gaussian measures, μ and τ are shown separately; a mean reading time is μ + τ. Left: the status models, pooled over the two prequestion groups and both topics (prequestioned, non-prequestioned, and no-prequestion sentences). Other panels: the Group × Topic models, for the Polio and TB sentences in the Polio, TB, and no-prequestion groups; colours give the status of the sentences for the readers in that group. μ and τ are the Gaussian mean (typical reading time) and the tail mean (unusually long readings) of the ex-Gaussian models of selective regression-path duration (first-pass reading) and total gaze duration. Durations are in ms per word; regressions-in are the percentage of sentences returned to from later text; re-reading time is for sentences that were re-read. Preq. = prequestioned; non-preq. = non-prequestioned; no-preq. = no-prequestion. Points are posterior means; thick and thin bars are 66% and 95% credible intervals.

Discussion

This study used eye tracking to examine how prequestions influence the allocation of visual attention during reading and whether these changes are associated with subsequent memory benefits. On the final test, prequestions produced the expected specific benefit, with no reliable general benefit. Participants performed better on information related to their prequestioned topic, as reflected in the Group × Topic interaction (extreme evidence). Eye movements showed a similarly selective pattern. Compared with the same sentences read without prequestions, prequestioned sentences exhibited a greater contribution from unusually long reading times, including during first-pass reading. Readers also looked back to these sentences more frequently and spent more time rereading them after moving on to later material. In contrast, processing of non-prequestioned sentences did not reliably differ from the no-prequestion condition across the eye-movement measures. The Group × Topic interaction contrast for eye movements was positive (pd > 95%) for the tail of first-pass reading (τ), the tail

of total reading time (τ), look-backs from later text, and re-reading time; and not reliably positive for typical first-pass reading time (μ) and typical total reading time (μ) (Figure 5). Together, these findings suggest that prequestions selectively alter how readers process information relevant to the questions, rather than producing a general increase in processing across the passage.

The distributional analyses provide further insight into the nature of these changes. Prequestions did not produce a reliable shift in the Gaussian component (μ) of reading times, with Bayes factors favoring the absence of a difference in both first-pass reading and total gaze duration. Instead, prequestions increased the exponential component (τ), indicating a greater contribution from unusually long reading times. This pattern suggests that prequestions do not uniformly slow reading of relevant information, but instead alter the distribution in a way consistent with additional processing on some encounters with an answer. The increased frequency of look-backs and longer rereading times further indicate that readers devoted additional processing to prequestioned information after initially encountering it. Importantly, these changes were specific to the prequestioned sentences, with no corresponding increase or decrease in processing of non-prequestioned sentences. This specificity parallels the meta-analytic findings of King-Shepard et al. (2025) and St. Hilaire et al. (2024), who reported a robust specific memory benefit but neither a reliable general benefit nor a general cost. The absence of changes to non-prequestioned information also provides little support for accounts proposing that prequestions broadly increase attention or monitoring across an entire topic or passage.

One possible explanation for this pattern is that additional processing is triggered when readers recognize that a sentence contains the answer to a prequestion. Lewis and Mensink (2012) proposed a related mechanism in which information that overlaps with a prequestion in wording or meaning "resonates" with the reader's memory of the question, prompting additional

inspection. Such recognition may occur only after readers have processed enough of a sentence to identify its relevance. Moreover, recognition may not occur on every encounter, particularly if readers fail to notice the connection or no longer remember the question. Consequently, recognition-triggered processing could produce disproportionately long reading times on some occasions rather than a uniform increase in reading duration. This interpretation is consistent with the increased τ observed in the present study. Previous distributional research has similarly shown that some experimental manipulations primarily affect the right tail of response-time distributions rather than their location. For example, modest visual degradation of text has been found to shift μ (White & Staub, 2012), whereas familiarity in scene viewing has been associated with increases in τ (Guy et al., 2020). Although these distributional patterns do not uniquely identify cognitive processes, our findings are consistent with the possibility that recognizing question-relevant information sometimes triggers additional processing.

The timing of the eye-movement effects provides further insight into this possibility. The increase in τ during first-pass reading suggests that additional processing can occur during the initial encounter with an answer, before readers move on to subsequent text. At the same time, the greater frequency of look-backs and longer rereading times suggest that readers also return to relevant information after initially encountering it. These returns may provide opportunities to compare the information with the original prequestion or an earlier guess, resolve uncertainty, or rehearse the answer. Recognition of the connection between a prequestion and its answer may therefore play an important role in determining when readers allocate additional attention. Consistent with the importance of remembering prequestions, St. Hilaire and Carpenter (2020) found that prequestions enhanced learning from video lectures only when learners remembered the questions. The present findings are also relevant to the attentional-window account (Sana & Carpenter, 2023), which proposes that attention remains elevated while learners search for an answer and returns to baseline once the answer is found. Although this account can

accommodate increased processing while readers search for relevant information, it does not straightforwardly predict the additional look-backs and rereading observed after readers have already encountered the answer. These findings suggest that the influence of prequestions on visual processing may extend beyond the initial search for relevant information.

More broadly, the results suggest that the benefits of prequestions cannot be explained simply in terms of increased time spent attending to relevant information. If prequestions produced a uniform slowing of reading, we might expect a broader shift in reading-time distributions. Instead, the observed effects were concentrated in the right tail and in subsequent returns to the targeted sentences. This pattern is consistent with the possibility that prequestions influence the processing of relevant information after it has been recognized, rather than merely encouraging readers to allocate more attention to it in advance. Such an interpretation aligns with the *mental model hypothesis* proposed by Motz et al. (2026), according to which prequestions activate relevant prior knowledge that facilitates the processing of related information during learning. On this account, recognizing a connection between incoming information and previously activated knowledge may prompt additional attention or elaboration. However, the present study cannot determine whether such knowledge activation preceded the observed changes in eye movements. Furthermore, because the non-prequestioned sentences concerned a different disease, the design cannot establish whether activated knowledge would also facilitate processing of related but non-prequestioned information within the same topic.

The observed pattern is also compatible with metacognitive explanations of the prequestion effect. By revealing gaps in learners' knowledge, prequestions may encourage monitoring for relevant information during subsequent learning (King-Shepard et al., 2025). Recognizing that a sentence provides an answer may prompt learners to evaluate their earlier responses or attend more closely to information they previously could not retrieve. The longer readings and additional returns observed here are consistent with such monitoring, although eye

movements alone cannot distinguish metacognitive evaluation from elaboration, rehearsal, or other forms of additional processing. Future research combining eye tracking with measures such as judgments of learning or confidence ratings may help differentiate these possibilities (King-Shepard et al., 2025). Importantly, the present findings also do not establish that increased processing causes the specific memory benefit. Because reading behavior was measured rather than experimentally manipulated, the relationship between additional processing and subsequent memory remains correlational. Prequestions may also strengthen memory through processes that are not directly reflected in eye movements. Experiments that manipulate opportunities for rereading or returning to previously encountered answers could help determine whether these behaviors contribute causally to the prequestion effect.

The present findings also extend previous eye-tracking research on prequestions. Because participants in our TB prequestion condition read the same passage used in Experiment 2 of Lewis and Mensink (2012) and received prequestions about the same topic, comparisons between the TB and no-prequestion groups provide a close conceptual replication of that experiment. Consistent with Lewis and Mensink (2012), TB prequestions improved subsequent memory for TB information (11.8 percentage points, 95% CrI [5.3, 17.9]), increased the frequency of look-backs to TB sentences (10.0 percentage points, 95% CrI [3.0, 16.9]), and increased the time spent rereading those sentences (38.6 ms per word, 95% CrI [12.2, 67.1]). However, whereas Lewis and Mensink (2012) found more re-inspection of targeted sentences during first-pass reading, we found no reliable difference in first-pass reading of TB sentences when examining either μ (6.2 ms per word, 95% CrI [−13.9, 26.9]) or τ (7.6 ms per word, 95% CrI [−19.6, 35.3]). When data were pooled across both prequestion groups, first-pass reading showed an increase in τ without a corresponding shift in μ, although the evidence was weak.

Several methodological differences may contribute to the discrepancies between studies. The present study included a larger sample and counterbalanced the targeted content, allowing us

to compare reading of the same sentences with and without prequestions. We also measured rereading as time spent on a sentence after the reader had progressed beyond it, and operationalized first-pass processing using selective regression-path duration (Schotter & Dillon, 2025). This measure was particularly useful because approximately half of first passes were interrupted by brief returns to the preceding line. In addition, the distributional analyses allowed us to distinguish changes in the Gaussian and exponential components of reading times, which cannot be separated through analyses of means alone. This distinction is important because a difference in average reading time can arise from different changes in the underlying distribution, with potentially different implications for how prequestions influence reading.

Our findings also resemble those of Bostan and Ozcelik (2026), who observed more fixations on prequestioned sentences and more transitions between them. These patterns are broadly consistent with the increased look-backs and rereading observed here. However, unlike the present study and much of the prequestion literature, Bostan and Ozcelik (2026) also reported a general memory benefit. One possible explanation concerns the types of prequestions used. Their integrative questions required readers to combine information distributed across different parts of the passage, potentially encouraging processing of information beyond the sentences directly targeted by the questions (Pan & Carpenter, 2023). In contrast, the prequestions used in the present study targeted individual facts contained within single sentences, and neither memory nor eye movements showed reliable general benefits. This difference is consistent with the possibility that the scope of attentional processing depends on the information required to answer the prequestions. However, given the relatively small sample in Bostan and Ozcelik (2026) (12 participants per group), this explanation remains tentative.

Methodologically, these findings highlight the importance of selecting appropriate statistical analyses when examining eye-tracking data. Reading-time distributions are typically right-skewed, and experimental manipulations may affect different parts of these distributions in

different ways. Conventional analyses of mean reading times can identify overall differences between conditions but cannot distinguish a general shift in reading times from a change in the right tail of the distribution. In the present study, the ex-Gaussian analyses revealed that prequestions increased the exponential component (τ) of reading times without a corresponding shift in the Gaussian component (μ). This distinction would not have been apparent from analyses of means alone. Thus, distributional analyses can provide a more detailed characterization of how experimental manipulations influence reading behavior, revealing patterns that may otherwise be overlooked or obscured by conventional approaches.

One limitation of the present study concerns the characteristics of the reading materials. Each topic was represented by only five target sentences and five final-test questions. Moreover, the Polio and TB sentences differed in length, reading difficulty, content, and position within the passage. These differences are particularly important in eye-tracking research, where linguistic and visual characteristics of the text can influence fixation durations, regressions, and rereading independently of the experimental manipulation. Consequently, differences in eye movements between topics cannot necessarily be attributed to prequestions alone. To address this concern, our primary analyses compared the same sentences across conditions, allowing us to account for fixed characteristics of the sentences themselves. Nevertheless, the effects were not identical across topics. In particular, increases in look-backs and rereading were driven primarily by the TB prequestion group, whose target sentences also attracted more look-backs in the no-prequestion condition. These findings underscore the importance of carefully controlling text stimuli in eye-tracking research. Future studies should more closely match target sentences on relevant linguistic and visual characteristics, including sentence length, syntactic complexity, reading difficulty, and position within the passage, or systematically counterbalance which sentences are targeted by prequestions across versions of the same text.

In summary, prequestions selectively changed how readers processed information that answered the questions. Rather than producing a uniform slowing of reading, prequestions increased the contribution of unusually long reading times and prompted readers to return to relevant information more frequently and for longer periods. These effects were restricted to prequestioned sentences and accompanied a specific memory benefit, with no reliable general benefit. The pattern is consistent with the possibility that recognizing information as relevant to a prequestion triggers additional processing during initial reading and subsequent rereading. However, whether such processing contributes causally to improved memory, or instead accompanies other learning mechanisms initiated by prequestions, remains an open question.

References

Arel-Bundock, V., Greifer, N., & Heiss, A. (2024). How to interpret statistical models using marginaleffects for R and Python. *Journal of Statistical Software*, *111*(9), 1–32. https://doi.org/10.18637/jss.v111.i09

Ashby, J., Rayner, K., & Clifton, C. (2005). Eye movements of highly skilled and average readers: Differential effects of frequency and predictability. *The Quarterly Journal of Experimental Psychology Section A*, *58*(6), 1065–1086. https://doi.org/10.1080/02724980443000476

Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry: The power of response time distributional analyses. *Current Directions in Psychological Science*, *20*(3), 160–166. https://doi.org/10.1177/0963721411408885

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Benoit, K., Watanabe, K., Wang, H., Lua, J. W., & Kuha, J. (2024). *Quanteda.textstats: Textual statistics for the quantitative analysis of textual data*. https://quanteda.io

Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). Quanteda: An r package for the quantitative analysis of textual data. *Journal of Open Source Software*, *3*(30), 774. https://doi.org/10.21105/joss.00774

Berlyne, D. E. (1954). A theory of human curiosity. *British Journal of Psychology*, *45*(3), 180–191. https://doi.org/10.1111/j.2044-8295.1954.tb01243.x

Berlyne, D. E. (1962). Uncertainty and epistemic curiosity. *British Journal of Psychology*, *53*(1), 27–34. https://doi.org/10.1111/j.2044-8295.1962.tb00811.x

Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings. In J. Metcalfe & A. P. Shimamura (Eds.), *Metacognition: Knowing about knowing* (pp. 185–205). MIT Press.

Bjork, R. A., Dunlosky, J., & Kornell, N. (2013). Self-regulated learning: Beliefs, techniques, and illusions. *Annual Review of Psychology*, *64*, 417–444. https://doi.org/10.1146/annurev-psych-113011-143823

Bostan, E., & Ozcelik, E. (2026). The effect of integrative prequestions on learning from text: An eye-tracking study. *The Journal of Experimental Education*, *94*(1), 33–45. https://doi.org/10.1080/00220973.2024.2355980

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Carpenter, S. K. (2012). Testing enhances the transfer of learning. *Current Directions in Psychological Science*, *21*(5), 279–283. https://doi.org/10.1177/0963721412452728

Carpenter, S. K., King-Shepard, Q., & Nokes-Malach, T. J. (2023). The prequestion effect: Why it is useful to ask students questions before they learn. In C. E. Overson, C. M. Hakala, L. L. Kordonowy, & V. A. Benassi (Eds.), *In their own words: What scholars want you to know about why and how to apply the science of learning in your academic setting* (pp. 74–82). Society for the Teaching of Psychology.

Carpenter, S. K., Rahman, S., & Perkins, K. (2018). The effects of prequestions on classroom learning. *Journal of Experimental Psychology: Applied*, *24*(1), 34–42. https://doi.org/10.1037/xap0000145

Carpenter, S. K., & Toftness, A. R. (2017). The effect of prequestions on learning from video presentations. *Journal of Applied Research in Memory and Cognition*, *6*(1), 104–109. https://doi.org/10.1016/j.jarmac.2016.07.014

Carr, J. W., Pescuma, V. N., Furlan, M., Ktori, M., & Crepaldi, D. (2022). Algorithms for the automated correction of vertical drift in eye-tracking data. *Behavior Research Methods*, *54*(1), 287–310. https://doi.org/10.3758/s13428-021-01554-0

Daneman, M., & Carpenter, P. A. (1980). Individual differences in working memory and reading. *Journal of Verbal Learning and Verbal Behavior*, *19*(4), 450–466. https://doi.org/10.1016/S0022-5371(80)90312-6

Deng, R., & Gao, Y. (2023). A review of eye tracking research on video-based learning. *Education and Information Technologies*, *28*(6), 7671–7702. https://doi.org/10.1007/s10639-022-11486-7

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, *14*(1), 4–58. https://doi.org/10.1177/1529100612453266

Firke, S. (2024). *Janitor: Simple tools for examining and cleaning dirty data*. https://github.com/sfirke/janitor

Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *Cmdstanr: R interface to CmdStan*. https://mc-stan.org/cmdstanr/

Gamer, M., Lemon, J., Fellows, I., & Singh, P. (2026). *Irr: Various coefficients of interrater reliability and agreement*. https://www.r-project.org

Geller, J., Carpenter, S. K., Lamm, M. H., Rahman, S., Armstrong, P. I., & Coffman, C. R. (2017). Prequestions do not enhance the benefits of retrieval in a STEM classroom.

Cognitive Research: Principles and Implications, *2*(1), 42. https://doi.org/10.1186/s41235-017-0078-z

Guy, N., Lancry-Dayan, O. C., & Pertzov, Y. (2020). Not all fixations are created equal: The benefits of using ex-Gaussian modeling of fixation durations. *Journal of Vision*, *20*(10), 9. https://doi.org/10.1167/jov.20.10.9

Hartwig, M. K., & Dunlosky, J. (2012). Study strategies of college students: Are self-testing and scheduling related to achievement? *Psychonomic Bulletin & Review*, *19*(1), 126–134. https://doi.org/10.3758/s13423-011-0181-y

Hyönä, J., Radach, R., & Deubel, H. (Eds.). (2003). *The mind's eye: Cognitive and applied aspects of eye movement research*. Elsevier.

Kallioinen, N., Paananen, T., Bürkner, P.-C., & Vehtari, A. (2023). Detecting and diagnosing prior and likelihood sensitivity with power-scaling. *Statistics and Computing*, *34*. https://doi.org/10.1007/s11222-023-10366-5

Kay, M. (2024). ggdist: Visualizations of distributions and uncertainty in the grammar of graphics. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 414–424. https://doi.org/10.1109/TVCG.2023.3327195

Kincaid, J. P., Fishburne, R. P., Rogers, R. L., & Chissom, B. S. (1975). *Derivation of new readability formulas (automated readability index, fog count and Flesch reading ease formula) for Navy enlisted personnel* (Nos. Research Branch Report 8-75). Naval Technical Training Command.

King-Shepard, Q. W., Walker, J., Nokes-Malach, T. J., Carpenter, S. K., & Fraundorf, S. H. (2025). The effect of prequestions on learning: A multilevel meta-analysis. *Educational Psychology Review*, *37*, 115. https://doi.org/10.1007/s10648-025-10075-7

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, *15*(2), 155–163. https://doi.org/10.1016/j.jcm.2016.02.012

Kornell, N., Hays, M. J., & Bjork, R. A. (2009). Unsuccessful retrieval attempts enhance subsequent learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *35*(4), 989–998. https://doi.org/10.1037/a0015729

Kornell, N., & Vaughn, K. E. (2016). How retrieval attempts affect learning: A review and synthesis. In B. H. Ross (Ed.), *Psychology of learning and motivation* (Vol. 65, pp. 183–215). Academic Press. https://doi.org/10.1016/bs.plm.2016.03.003

Kumle, L., Võ, M. L.-H., & Draschkow, D. (2021). Estimating power in (generalized) linear mixed models: An open introduction and tutorial in R. *Behavior Research Methods*, *53*(6), 2528–2543. https://doi.org/10.3758/s13428-021-01546-0

Lee, M. D., & Wagenmakers, E.-J. (2014). *Bayesian cognitive modeling: A practical course*. Cambridge University Press. https://doi.org/10.1017/CBO9781139087759

Lewis, M. R., & Mensink, M. C. (2012). Prereading questions and online text comprehension. *Discourse Processes*, *49*(5), 367–390. https://doi.org/10.1080/0163853X.2012.662801

Liu, H. (2008). Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science*, *9*(2), 159–191.

Lo, S., & Andrews, S. (2015). To transform or not to transform: Using generalized linear mixed models to analyse reaction time data. *Frontiers in Psychology*, *6*, 1171. https://doi.org/10.3389/fpsyg.2015.01171

Lüdecke, D., Waggoner, P., & Makowski, D. (2019). insight: A unified interface to access information from model objects in R. *Journal of Open Source Software*, *4*(38), 1412. https://doi.org/10.21105/joss.01412

Madsen, J., Júlio, S. U., Gucik, P. J., Steinberg, R., & Parra, L. C. (2021). Synchronized eye movements predict test scores in online video education. *Proceedings of the National Academy of Sciences*, *118*(5), e2016980118. https://doi.org/10.1073/pnas.2016980118

Makowski, D. (2026). *Cogmod: Cognitive models for subjective scales and decision making tasks*. https://github.com/DominiqueMakowski/cogmod

Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the Bayesian framework. *Frontiers in Psychology*, *10*, 2767. https://doi.org/10.3389/fpsyg.2019.02767

Makowski, D., Ben-Shachar, M. S., & Lüdecke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the bayesian framework. *Journal of Open Source Software*, *4*(40), 1541. https://doi.org/10.21105/joss.01541

Makowski, D., Ben-Shachar, M. S., Wiernik, B. M., Patil, I., Thériault, R., & Lüdecke, D. (2025). modelbased: An R package to make the most out of your statistical models through marginal means, marginal effects, and model predictions. *Journal of Open Source Software*, *10*(109), 7969. https://doi.org/10.21105/joss.07969

Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-Gaussian and shifted Wald parameters: A diffusion model analysis. *Psychonomic Bulletin & Review*, *16*(5), 798–817. https://doi.org/10.3758/PBR.16.5.798

McNamara, D. S., & Magliano, J. (2009). Toward a comprehensive model of comprehension. In B. H. Ross (Ed.), *The psychology of learning and motivation* (Vol. 51, pp. 297–384). Elsevier. https://doi.org/10.1016/S0079-7421(09)51009-2

Motz, B. A., Chinni, A., de Leeuw, J. R., Jankowski, H. E., Aggarwal, A., Amato, M., Berlin, K., Britten, K., Brown, A., Cerchiaro, M. L., Evans, N. J., Findley, A., Gorman, R., Gregg, K. D., Hansen, K., Hollender, H., Hullinger, R. A., Larkin, P., Lion, M., … Fyfe, E. R. (2026). ManyClasses 2: The effects of prequestions on media interactions and learning.

Journal of Educational Psychology, *118*(2), 153–175. https://doi.org/10.1037/edu0000978

Olsen, A. (2012). *The Tobii I-VT fixation filter: Algorithm description*. Tobii Technology.

Pan, S. C., & Carpenter, S. K. (2023). Prequestioning and pretesting effects: A review of empirical research, theoretical perspectives, and implications for educational practice. *Educational Psychology Review*, *35*, 97. https://doi.org/10.1007/s10648-023-09814-5

Pedersen, T. L. (2025). *Patchwork: The composer of plots*. https://patchwork.data-imaginist.com

Perfetti, C. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading*, *11*(4), 357–383. https://doi.org/10.1080/10888430701530730

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://doi.org/10.32614/R.manuals

Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, *124*(3), 372–422. https://doi.org/10.1037/0033-2909.124.3.372

Rayner, K. (2009). Eye movements and attention in reading, scene perception, and visual search. *Quarterly Journal of Experimental Psychology*, *62*(8), 1457–1506. https://doi.org/10.1080/17470210902816461

Richland, L. E., Kornell, N., & Kao, L. S. (2009). The pretesting effect: Do unsuccessful retrieval attempts enhance learning? *Journal of Experimental Psychology: Applied*, *15*(3), 243–257. https://doi.org/10.1037/a0016496

Rivers, M. L., Berdelis, A. J., & Pan, S. C. (2026). Does covert prequestioning enhance learning? Implications for the roles of attention and retrieval in the prequestioning effect. *Applied Cognitive Psychology*. https://doi.org/10.1002/acp.70165

Roediger, H. L., & Karpicke, J. D. (2006). The power of testing memory: Basic research and implications for educational practice. *Perspectives on Psychological Science*, *1*(3), 181–210. https://doi.org/10.1111/j.1745-6916.2006.00012.x

Rothkopf, E. Z., & Billington, M. J. (1979). Goal-guided learning from text: Inferring a descriptive processing model from inspection times and eye movements. *Journal of Educational Psychology*, *71*(3), 310–327. https://doi.org/10.1037/0022-0663.71.3.310

Rowland, C. A. (2014). The effect of testing versus restudy on retention: A meta-analytic review of the testing effect. *Psychological Bulletin*, *140*(6), 1432–1463. https://doi.org/10.1037/a0037559

Salvucci, D. D., & Goldberg, J. H. (2000). Identifying fixations and saccades in eye-tracking protocols. *Proceedings of the 2000 Symposium on Eye Tracking Research & Applications*, 71–78. https://doi.org/10.1145/355017.355028

Sana, F., & Carpenter, S. K. (2023). Broader benefits of the pretesting effect: Placement matters. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-023-02274-6

Sana, F., Forrin, N. D., Sharma, M., Dubljevic, T., Ho, P., Jalil, E., & Kim, J. A. (2020). Optimizing the efficacy of learning objectives through pretests. *CBE—Life Sciences Education*, *19*(3), ar43. https://doi.org/10.1187/cbe.19-11-0257

Schotter, E. R., & Dillon, B. (2025). A beginner's guide to eye tracking for psycholinguistic studies of reading. *Behavior Research Methods*, *57*(2), 68. https://doi.org/10.3758/s13428-024-02572-4

Serrano-Carot, M., Angele, B., Xu, H., & Vasilev, M. R. (2025). *Webcams can be used to study eye movements during reading*. PsyArXiv preprint; PsyArXiv. https://doi.org/10.31234/osf.io/bzt2h_v1

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, *86*(2), 420–428. https://doi.org/10.1037/0033-2909.86.2.420

Silveira, N., Dozat, T., Marneffe, M.-C. de, Bowman, S., Connor, M., Bauer, J., & Manning, C. (2014). A gold standard dependency corpus for English. *Proceedings of the Ninth*

International Conference on Language Resources and Evaluation (LREC'14), 2897–2904.

Soderstrom, N. C., & Bjork, E. L. (2023). Pretesting enhances learning in the classroom. *Educational Psychology Review*, *35*(3), 88. https://doi.org/10.1007/s10648-023-09805-6

St. Hilaire, K. J., & Carpenter, S. K. (2020). Prequestions enhance learning, but only when they are remembered. *Journal of Experimental Psychology: Applied*, *26*(4), 705–716. https://doi.org/10.1037/xap0000296

St. Hilaire, K. J., Carpenter, S. K., & Jennings, J. M. (2019). Using prequestions to enhance learning from reading passages: The roles of question type and structure building ability. *Memory*, *27*(9), 1204–1213. https://doi.org/10.1080/09658211.2019.1641209

St. Hilaire, K. J., Chan, J. C. K., & Ahn, D. (2024). Guessing as a learning intervention: A meta-analytic review of the prequestion effect. *Psychonomic Bulletin & Review*, *31*(2), 411–441. https://doi.org/10.3758/s13423-023-02353-8

Staub, A., White, S. J., Drieghe, D., Hollway, E. C., & Rayner, K. (2010). Distributional effects of word frequency on eye fixation durations. *Journal of Experimental Psychology: Human Perception and Performance*, *36*(5), 1280–1293. https://doi.org/10.1037/a0016896

Straka, M., & Straková, J. (2017). Tokenizing, POS tagging, lemmatizing and parsing UD 2.0 with UDPipe. *Proceedings of the CoNLL 2017 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies*, 88–99. https://doi.org/10.18653/v1/K17-3009

Urbanek, S. (2026). *Png: Read and write PNG images*. https://www.rforge.net/png/

Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage–Dickey method. *Cognitive Psychology*, *60*(3), 158–189. https://doi.org/10.1016/j.cogpsych.2009.12.001

White, S. J., & Staub, A. (2012). The distribution of fixation durations during reading: Effects of stimulus quality. *Journal of Experimental Psychology: Human Perception and Performance*, *38*(3), 603–617. https://doi.org/10.1037/a0025338

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., Pedersen, T. L., & Seidel, D. (2025). *Scales: Scale functions for visualization*. https://scales.r-lib.org

Wijffels, J., & Straka, M. (2026). *Udpipe: Tokenization, parts of speech tagging, lemmatization and dependency parsing with the UDPipe 'NLP' toolkit*. https://bnosac.github.io/udpipe/en/index.html

Xie, Y. (2015). *Dynamic documents with R and knitr* (2nd ed.). Chapman; Hall/CRC. https://yihui.org/knitr/

Xie, Y. (2026). *Xfun: Supporting functions for packages maintained by yihui xie*. https://github.com/yihui/xfun

Yang, J., Zhang, Y., Pi, Z., & Xie, Y. (2021). Students' achievement motivation moderates the effects of interpolated pre-questions on attention and learning from video lectures. *Learning and Individual Differences*, *91*, 102055. https://doi.org/10.1016/j.lindif.2021.102055

Appendix A

Questions and Scoring Rules

Prequestion and final test responses were scored with the same rule-based script (`scoring_rubric.R`), written by Claude Opus 5.5 (Anthropic; model ID `claude-opus-5-5`) and

applied identically to every response. Each question awarded one point per correct element present in the response (Table A1, Table A2). Blank responses, "I don't know" and similar responses, and random keystrokes scored 0. Minor misspellings were accepted when the intended word was unambiguous (e.g., "afganistan"). The same rules were used for the prequestions and the final test.

Table A1

Final Test Questions and Answers

Q	Topic	Question	Answer	Points
1	Polio	What percent of the global budget does the Gates Foundation devote toward the eradication of polio	17%	1
2	Polio	Is there a cure for polio?	no	1
3	Polio	In which countries is polio still a major concern?	Nigeria, India, Pakistan, and Afghanistan	4
4	Polio	What measures are included in the modern treatment of polio?	antibiotics to prevent infections in weakened muscles, analgesics for pain, moderate exercise, and a nutritious diet	4
5	TB	What are the symptoms of tuberculosis?	a chronic cough with blood-tinged sputum, fever, night sweats, and weight loss	4
6	TB	What percentage of new infections will turn into active cases of tuberculosis?	10%	1
7	TB	What is the treatment for tuberculosis?	a long-term course of two antibiotics to kill the bacteria.	3
8	TB	Can people with latent tuberculosis spread the disease?	no	1
9	TB	How is tuberculosis spread through the air?	cough, sneeze, or spit	3
10	Polio	Since the development of the polio vaccine, the global number of polio cases per year has dropped from _____ to _____.	many hundreds of thousands to around a thousand	2

Note. Points = maximum points for the question. The Polio prequestion condition answered questions 1–4 and 10 before reading; the TB prequestion condition answered questions 5–9.

Table A2

Scoring Rules and Agreement With the Second Rater for Each Question

Q	Credited elements	*n*	Exact	ICC
1	The number 17 (as a whole number, e.g., not 170).	158	100.0%	1.00
2	A negative answer: “no”, “no cure”, or “not a cure”.	198	100.0%	1.00
3	One point each for Nigeria, India, Pakistan, and Afghanistan.	164	97.0%	.99
4	One point each for antibiotics; analgesics or pain relief; exercise (including physical activity or therapy); and diet or nutrition. “Antibiotic resistance” alone was not credited.	162	98.8%	.99
5	One point each for cough (or blood-tinged sputum/ phlegm); fever (or temperature); sweats (including night sweats); and weight loss.	180	91.7%	.96
6	10% (or “ten percent”, 1/10); not credited if stated as “less than” or “under” 10%.	155	99.4%	.99
7	One point each for antibiotics (or antibacterials); two (two drugs, courses, rounds, or types); and long-term (long, extended, prolonged, or months).	168	97.6%	.98
8	A negative answer (“no”).	185	100.0%	1.00
9	One point each for cough, sneeze, and spit (or saliva).	174	99.4%	1.00
10	One point each for a starting value of hundreds of thousands (100,000–999,999) and an end value of about a thousand (500–9,999, or “a thousand”).	134	91.0%	.94

Note. One point was awarded for each credited element present in a response, up to the question’s maximum (see Table A1). *n* = number of non-blank responses (prequestions and final

test) scored by both the automated rubric and the second rater; Exact = percentage of responses given the same score; ICC = ICC(2,1), absolute agreement.

Appendix B

Model Specifications

Table B1

Bayesian Models: Families, Effects, and Priors

Measure	Family	Random effects	Prior on comparisons
Total gaze duration	Ex-Gaussian (μ identity; σ, τ softplus)	μ and τ: participants (intercept, Topic slope), sentences (intercept, Status or Group slope); σ: participants (intercept)	μ: normal(0, 0.1) s/word; τ: normal(0, 0.5)
Selective regression-path duration	Ex-Gaussian (μ identity; σ, τ softplus)	μ and τ: participants (intercept, Topic slope), sentences (intercept, Status or Group slope); σ: participants (intercept)	μ: normal(0, 0.1) s/word; τ: normal(0, 0.5)
Re-reading time	Gamma (log)	Participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	normal(0, 0.5)
Regression-in (look-back)	Bernoulli (logit)	Participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	normal(0, 1.5)
Final test	Beta-binomial (logit)	Participants (intercept, Topic slope), questions (intercept, Status or Group slope)	normal(0, 1.5)
Prequestions	Beta-binomial (logit)	Participants (intercept), questions (intercept)	normal(0, 1.5)

Note. Pooled models: Status (prequestioned, non-prequestioned, no-prequestion) + Topic; Group × Topic models: Group (Polio, TB, no-prequestion) × Topic. Status, Condition, and Topic were coded with equal-prior contrasts, so the prior on the coefficients is also the prior on every pairwise difference between conditions (on the link scale). Other priors: intercepts normal(−1.5, 1) on the log scale (Gamma models; about 0.2 s per word), normal(0, 1.5) on the log-odds scale, and normal(0.2, 0.2) s per word for μ; ex-Gaussian σ and τ intercepts normal(−2.5, 1) on the softplus scale; random-effect *SD*s Student-*t*(3, 0, *ν*) with *ν* = 0.5 (Gamma models, ex-Gaussian σ and τ), 1 (logistic and beta-binomial models), or 0.1 (ex-Gaussian μ); random-effect correlations LKJ(2); the Gamma shape gamma(2, 0.5); beta-binomial overdispersion exponential(0.1). Durations were modelled in s per word.

Appendix C

Deviations From the Preregistration

Table C1

Deviations From the Preregistration and Their Effect on the Results

Aspect	Preregistered	As conducted	Reason	Effect on results
Prior-knowledge exclusion	Exclude on prior knowledge (no criterion)	Not applied	No criterion specified; self-rated familiarity and prequestion scores were low	Minor
Sample size	135 (45 per condition)	138 (47/45/46)	Exclusions left conditions unequal	Minor
DV standardized by length	Not specified	Durations in ms per word	Target sentences differ in length	Minor: pooled comparisons use the same sentences
Regression measure	First-pass regressions	Regressions-in (look-backs)	Error in the preregistration	None: intended measure reported
Inference	$p < .05$ (sidedness not specified)	Bayesian; pd > 95% (97.5% for general effects); Bayes factors	Bayes factors can support the null	Minor: pd > 95% ≈ one-sided $\alpha = .05$
Random effects	Participant intercepts and Topic slopes; some item intercepts	Maximal: plus item intercepts and Status/Group slopes	Maximal structure (Barr et al., 2013)	Minor; conservative

Aspect	Preregistered	As conducted	Reason	Effect on results
Ex-Gaussian σ	Group × Topic + participant effects	Participant intercepts only	σ not of interest	Minor: σ not interpreted
First-pass measure	First-pass duration	Selective regression-path duration	About half of first passes were cut short by a brief return to the preceding line, so first-pass duration was bimodal and no distribution described it	Minor: both measure reading before the reader moves on to later text; selective regression-path duration is not cut short by brief returns
Re-reading model	Linear or ex-Gaussian	Gamma	Re-reading times were close to exponential, so the ex-Gaussian's Gaussian component collapsed	Minor
Final test and prequestion scores	Points per topic, binomial	Points per question, beta-binomial	Allows question random effects	Minor

Aspect	Preregistered	As conducted	Reason	Effect on results
Specific and general effects	Group × Topic; prequestions vs. control	Status (prequestioned, non-prequestioned, control) + Topic	Separates specific and general effects	None: Group × Topic also reported

Note. Effect on results: *minor* = unlikely to change any conclusion. Sample sizes are no prequestions/Polio/TB. None of the preregistered alternatives was analysed, so the effects are judged from the nature of each change rather than estimated.

Appendix D

Drift Correction

Each fixation was assigned to a text line after correcting for vertical drift, separately for each reader and page. The correction combines two approaches used in automated drift-correction algorithms (Carr et al., 2022): assigning sequences of fixations to a line together, and modelling drift as a function of position on the page. First, a single vertical offset was estimated: candidate shifts of the recorded positions were tried in steps of 0.05 line spacings, and the shift that minimized the summed squared distance between each fixation and its nearest text line was kept (distances were capped at one line spacing, so that stray fixations in the margins had little influence). Second, consecutive fixations moving along a line were grouped into chains; a chain ended at a horizontal jump larger than 10% of the screen width (e.g., a return sweep to the next line) or a vertical jump larger than 0.6 line spacings. Each chain was assigned as a whole to the text line nearest its median position, so that a single fixation drifting toward a neighbouring line stayed on the line being read. Third, because drift often grew down the page or across it, a linear model of the recorded vertical position as a function of the assigned line's position and the fixation's horizontal position (recorded $y = a + b \times \text{line } y + c \times x$) was fitted to fixations whose chain lay within half a line spacing of its line; gaze positions were corrected by inverting this model and the chains were reassigned, and this was repeated five times. The linear model was used only if it was plausible (a vertical stretch within 20% and a tilt of less than one line across the screen); otherwise the single offset was kept. Throughout, the correction was limited everywhere on the page to the calibration accuracy criterion of 0.5° (0.57 cm, about 0.45 of a line spacing), so it could not move a reader's gaze by a whole line. Chains more than half a line spacing from every text line were treated as off the text.